Electric Vehicle Market Performance & Insights

Project Report

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1. Introduction

1.1 Project Overview

This project analyses the electric vehicle (EV) market using the Electric Vehicle Market Classification Dataset from Kaggle (file: EV Vehicle.csv.gz; stored in the GitHub repository as two compressed files, compressed_data.csv.gz and compressed_data-2.csv.gz). The data covers manufacturers, models, model year, battery type and capacity, driving range, price, customer rating, annual sales, market region, manufacturing country and CO2 saved per year. The analysis is presented as an interactive Tableau dashboard and story, supported by a Python Flask web application and a static project website published with GitHub Pages that embeds the Tableau Public dashboard and story.

Dataset: EV Vehicle.csv.gz

Kaggle link: https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset

1.2 Purpose

As a student project, the purpose is to practise the full data analytics workflow: understanding a real-world dataset, cleaning it, building visualizations, publishing the result online and documenting it. The specific goals are:

- Turn a large EV dataset into clear visuals that a non-technical viewer can read quickly.
- Find which manufacturers, battery types, regions and vehicle categories lead in sales.
- Compare customer ratings, driving range and battery capacity across vehicle categories.
- Show where EVs are manufactured and sold using a map and regional charts.
- Learn to publish a dashboard on Tableau Public and share the project through GitHub.

2. Ideation Phase

2.1 Problem Statement

The EV market dataset has many columns and a very large number of records, so it is difficult to see which manufacturers, battery technologies and regions perform best by reading the raw table. Students, analysts and business users need one filterable view that summarises sales, ratings and range in a simple way.

2.2 Empathy Map Canvas

Quadrant	Notes
Says	"Which manufacturers and regions sell the most EVs?"
Thinks	The raw data is too large; I need a simple summary I can trust.
Does	Scrolls through the spreadsheet and compares values manually.
Feels	Overwhelmed by numbers; wants quick and clear answers.
Pains	Many columns, no filters, slow manual analysis.
Gains	One dashboard with KPIs, filters, charts and a guided story.

2.3 Brainstorming

- KPI strip showing manufacturers, EVs, total sales, average rating, average range and average battery capacity.
- Seven worksheets: Manufacturer Sales, EV Sales by Battery Type, Regional Sales, Vehicle Sales by Category, Customer Rating, Range Category and Manufacturing Map.
- Filters on the dashboard for manufacturer and battery type.
- A Tableau Story (nine story points) that explains the market step by step.
- A Flask app and a GitHub project website to share the work.

3. Requirement Analysis

3.1 Customer Journey Map

Awareness	Exploration	Analysis	Decision	Follow-up
Opens the dashboard or website link	Uses manufacturer and battery filters	Reads the charts, map and story	Identifies leading manufacturers, regions and battery types	Shares the link or report

3.2 Solution Requirement

Type	Requirement	Detail
Functional	Interactive dashboard	KPI strip, seven charts and filters
Functional	Data preparation	Load the CSV, remove duplicates, handle missing values
Functional	Story	Nine story points from market snapshot to executive summary
Functional	Publishing	Publish to Tableau Public and GitHub
Non-functional	Usability	Clear colour theme, titles, labels and tooltips
Non-functional	Performance	Dashboard responds quickly when filters change

3.3 Data Flow Diagram

Kaggle dataset (EV Vehicle.csv.gz, stored in the repository as compressed_data.csv.gz and compressed_data-2.csv.gz) -> data preparation (duplicates removed, missing values handled) -> Tableau workbook (.twbx) with embedded extract -> Tableau Public (dashboard and story) ->

GitHub repository and project website (index.html on GitHub Pages) -> end user. The Flask app (app.py) reads the same compressed data files and serves the data to a local web dashboard; GitHub Pages hosts only static files, so the Flask app is run locally.

[Insert data flow diagram image here]

3.4 Technology Stack

Layer	Technology
Data	Kaggle Electric Vehicle Market Classification Dataset (CSV, gzip)
Visualization	Tableau Desktop and Tableau Public
Backend	Python, Flask, pandas
Frontend	HTML, CSS, JavaScript
Hosting / Version control	GitHub and GitHub Pages

4. Project Design

4.1 Problem Solution Fit

The problem of large, hard-to-read EV data is solved with one interactive dashboard that combines KPIs, filters, charts and a map. Users reach insights in a few clicks instead of analysing the raw table.

4.2 Proposed Solution

- Download and prepare the dataset (EV Vehicle.csv.gz), store it in the repository as two compressed files (compressed_data.csv.gz and compressed_data-2.csv.gz) and keep the original data unchanged.
- Build seven worksheets in Tableau and combine them into the EV Executive Dashboard, Executive Overview and EV Market Journey dashboards.
- Create a Tableau Story with nine story points and publish it on Tableau Public.
- Develop a Flask app (app.py), build a static project website (index.html) published with GitHub Pages, and upload all project files to GitHub.

4.3 Solution Architecture

Data layer: compressed CSV files. Processing layer: Tableau data extract and pandas in the Flask app. Presentation layer: Tableau Public dashboard and story, plus the Flask web page. Users open the project through a web browser using the Tableau Public and GitHub links.

[Insert architecture diagram image here]

5. Project Planning & Scheduling

5.1 Project Planning

Task No.	Task	Duration	Status
1	Dataset study, download and cleaning	4 days	Completed
2	Tableau worksheets and calculated fields	4 days	Completed
3	Dashboard design, filters and story creation	4 days	Completed
4	Publishing on Tableau Public and fixing publishing problems	4 days	Completed
5	Flask app, GitHub upload and final report	4 days	Completed

Problems Faced During Development

The biggest problem was publishing the dashboard. I struggled a lot to find the correct Tableau server to publish to. Tableau Desktop offers different publish options (Tableau Server, Tableau Cloud and Tableau Public), and a normal Tableau Server or Cloud address was not available to me as a student. After a lot of searching I understood that the dashboard has to be published through Tableau Public using a free Tableau Public account, which is the correct place for a student project. The workbook also had to use a saved extract, because Tableau Public does not allow live connections to local files. Once this was clear, the dashboard and story were published successfully and the links were added to the GitHub repository.

6. Functional and Performance Testing

6.1 Performance Testing

Test	Description	Result
Filter test	Manufacturer and battery type filters update the charts	Working
KPI check	KPI values on the dashboard are consistent with the data	Working
Publishing test	Dashboard and story open from the Tableau Public links	Working
Link test	GitHub repository and project links open correctly	Working
Flask app test	app.py runs locally and shows the dashboard page	[Confirm after running]

7. Results

7.1 Output Screenshots

Dashboard views: EV Executive Dashboard, Executive Overview and EV Market Journey. The KPI strip on the dashboard shows 200 manufacturers, 17,980 EVs, 1.55B total sales, an average

customer rating of 4.09/5, an average range of 324.5 km and an average battery capacity of 78.2 kWh.

- [Screenshot 1: EV Executive Dashboard]
- [Screenshot 2: Tableau Story]
- [Screenshot 3: Manufacturing Map worksheet]
- [Screenshot 4: Flask app running locally]
- [Screenshot 5: GitHub repository]

Live links:

- Tableau Dashboard: https://public.tableau.com/app/profile/mausam.kumar.pandit/viz/EV_Market_Performance_Market_Story_2026-10-25/EVExecutiveDashboard
- Tableau Story: https://public.tableau.com/app/profile/mausam.kumar.pandit/viz/EV_Market_Performance_Green_Executive_Refresh/Story2
- GitHub Repository: https://github.com/mausam9039208144-dev/EV-Market-Performance-Insights
- Project Website (GitHub Pages): https://mausam9039208144-dev.github.io/EV-Market-Performance-Insights/
- Dataset (Kaggle): https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset

8. Advantages & Disadvantages

Advantages

- Interactive and easy to understand, even for non-technical users.
- Filters, KPIs and a map give a quick overview of the EV market.
- Published online, so it can be shared with a single link.
- All project files are available on GitHub.

Disadvantages

- Tableau Public makes the data extract public, so only shareable data can be used.
- Publishing needs a Tableau Public account and an internet connection.
- The dataset total values should be checked before treating them as real market-wide figures.
- The Flask app runs locally unless a Python hosting service is used.

9. Conclusion

The project converts a large EV market dataset into a clear dashboard and story that show sales, battery technology, regional performance, customer ratings and driving range. It also taught me the complete process of preparing data, building visuals in Tableau, solving the publishing problem on Tableau Public and sharing the work through GitHub.

10. Future Scope

- Add forecasting of future EV sales.
- Use machine learning to classify EV market segments from the dataset.
- Host the Flask app online with a database.
- Add more data such as charging stations and price trends.

11. Appendix

Source Code

All project files are available in the GitHub repository: https://github.com/mausam9039208144-dev/EV-Market-Performance-Insights. The repository contains:

- EV_Market_Performance_Market_Story.twbx: Tableau workbook with the dashboards, filters and story.
- app.py: Flask application that processes the data and serves the local dashboard.
- index.html: static project website published with GitHub Pages.
- compressed_data.csv.gz and compressed_data-2.csv.gz: compressed dataset, parts 1 and 2.
- EV_Market_Performance_Final_Report-3.docx: this project report.
- README.md: project documentation, including how to run the Flask app locally.

Dataset Link

EV Vehicle.csv.gz (Kaggle): https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset

GitHub & Project Demo Link

- GitHub: https://github.com/mausam9039208144-dev/EV-Market-Performance-Insights
- Project Website (GitHub Pages): https://mausam9039208144-dev.github.io/EV-Market-Performance-Insights/
- Tableau Dashboard: https://public.tableau.com/app/profile/mausam.kumar.pandit/viz/EV_Market_Performance_Market_Story_2026-10-25/EVExecutiveDashboard
- Tableau Story: https://public.tableau.com/app/profile/mausam.kumar.pandit/viz/EV_Market_Performance_Green_Executive_Refresh/Story2